

Gravity, Matter, and Abelian Gauge Terminals

from One Ordered Tau Descent

Tau Core Foundation Paper VII: First Proof Packet

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Abstract

We ask whether the already constructed Tau gravity, temporal, quantum, matter-occupation and Abelian gauge terminals can belong to one physical descent rather than to independently fitted sector models. The inherited results do not yet prove that Nature selects such a descent. They do, however, permit a finite conditional construction. We identify the joint post-body object with one renormalized in-in effective functional on each stationary body fiber. Its source, metric, gauge and clock-character derivatives generate probabilities, stress, current and clock response, respectively. Green operators are inverse Hessians rather than additional action terms, and finite local counterterms have one ledger entry even when their placement convention is changed. A disjoint occupation ledger assigns every physical mode to exactly one classical or quantum stress export. Under diffeomorphism and $U(1)$ gauge invariance, stationarity gives the semiclassical Einstein–Maxwell–matter equations and their Ward identities. Their weak field limit is the Poisson equation sourced by the same once-counted total energy density, together with the ordinary Maxwell equations. We prove that a simultaneous body/post-body Hessian cannot retain nonzero body conditioning and exact no-backreaction, and that duplicate classical/quantum occupation necessarily double counts stress. Conversely, recovering standard terminal equations does not identify common Tau provenance: an equal-terminal independent-source counterfamily remains. The result is therefore a conditional source-assembly and non-double-counting theorem, not a derivation of quantum gravity, the Standard Model, physical constants or Nature-level source selection.

Inputs from earlier papers. Paper I supplies the body-first descent; Paper II the typed source support; Paper III recoverable record transport; Paper IV the parent-selection boundary; Paper V the observer clock and causal leaf; Paper VI the conditional quantum terminal. The theory hub supplies the P1 gravity–time–radiation ordered ledger and the P3 Abelian handoff.

Results owned here. The joint terminal ledger, the disjoint occupation rule, the effective-source derivative ledger, the Ward/Bianchi consistency theorem, the simultaneous-action and duplicate-stress no-go results, and the same-reduct Nature-selection no-go.

1 Question and claim boundary

Let M_τ be the stabilized atemporal morphological body. Paper VII asks:

$$\boxed{\begin{array}{l} \text{Can gravity, matter, gauge/radiation and quantum backreaction} \\ \text{descend from one once-counted ordered physical source ledger?} \end{array}} \quad (1)$$

The word *one* denotes common provenance and calibration. It does not mean one emitter, one scalar varied at every generative stage, or one terminal observable. The generative order remains

$$B_\tau^{\text{phys}} + s_U \longrightarrow M_\tau \longrightarrow \mathfrak{L}_7 \longrightarrow \text{Desc}_O \longrightarrow \{g, A, \Phi, \rho_O\}. \quad (2)$$

No terminal residual may select M_τ or the ledger.

2 Inherited-results ledger

Result	Owner	Status and assumptions	Use here
Body-first descent	Paper I	conditional physical MVP	generative order
Typed incidence	Paper II	finite source theorem	occupied domains
Record recovery	Paper III	exact/approximate conditional theorem	observer terminal
Parent selection	Paper IV	present reduct does not select full packet	claim boundary
Temporal descent	Paper V	conditional clock/causal leaf	time and weak limit
Quantum terminal	Paper VI	conditional complex/effect/QFT packet	Γ_q and ρ_O
P1 assembly	theory hub	conditional metric/time/radiation ledger	g , null radiation, stress
P3 Abel handoff	theory hub	conditional $Z_F = p_* K_Y$	Maxwell/current sector

Table 1: Dependency and ownership ledger. No inherited conditional result is promoted to a Nature theorem.

3 The ordered Paper-VII ledger

Let y denote body coordinates, and write $z = (g, A, \Phi, \rho)$ for post-body geometry, Abelian gauge, occupied classical matter and quantum-state data.

Definition 1 (Once-counted ordered ledger). *The Paper-VII ledger is*

$$\mathfrak{L}_7 = (\mathcal{A}_{\text{pre}}[y; B, s], y_*, \Gamma_7[g, A, \Phi, \rho \mid y_*], \text{Desc}_{O \leftarrow y_*}, \mathcal{U}), \quad (3)$$

where $D_y \mathcal{A}_{\text{pre}}(y_*) = 0$, the physical body Hessian is positive, and

$$\Gamma_7 = \Gamma_{\text{grav}}[g \mid y_*] + \Gamma_{\text{EM}}[A, g \mid y_*] + \Gamma_{\text{occ}}[\Phi, A, g \mid y_*] + \Gamma_q[\rho, A, g \mid y_*] + \Gamma_{\text{ct}}[A, g]. \quad (4)$$

Each occupied mode and each counterterm appears in exactly one summand. Observer restriction is evaluated after stationarity and is not varied back against y_* . The unit packet $\mathcal{U}$ is shared by every summand.

4 One effective source on each stationary body fiber

The symbol Γ_7 need not be postulated as five unrelated pieces. Let φ collect the occupied quantum fields, let η_a couple to normalized observer effects, and let ϑ denote the occupied clock character. On the already selected stationary body y_* define the closed-time-path effective functional

$$\exp\left(\frac{i}{\hbar} W_O[g, A, \vartheta, \eta \mid y_*]\right) = \int_{\mathcal{C}_O} \mathcal{D}\varphi \exp\left[\frac{i}{\hbar} \Gamma_{\text{PC}}[y_*; g, A, \vartheta, \eta, \varphi]\right]. \quad (5)$$

The in-in contour is required for real causal expectation values. It is a terminal representation of an atemporal ordered functional fibration, not a parent-time history. Renormalization gives one W_O^{ren} whose derivatives define

$$p_a = \left. \frac{\delta W_O}{\delta \eta_a} \right|_{\eta=0}, \quad \langle T_{\mu\nu}^q \rangle_{\text{ren}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_O^{\text{ren}}}{\delta g^{\mu\nu}}, \quad (6)$$

$$\langle J_q^\mu \rangle_{\text{ren}} = \frac{1}{\sqrt{-g}} \frac{\delta W_O^{\text{ren}}}{\delta A_\mu}, \quad E_O = \frac{\delta W_O^{\text{ren}}}{\delta \vartheta}. \quad (7)$$

The final expression is schematic until the clock character and its normalization are fixed, but its ownership is unambiguous.

Theorem 1 (Effective common-source construction). *Assume the ordered pre-body solve, a source-owned probe-complete post-body functional Γ_{PC} , an observer-admissible integration class $\mathcal{C}_O$, and a local covariant renormalization prescription. Then Eqs. (5)–(7), together with the classical body and coherent mean-field terms, construct the Paper-VII Γ_7 ledger from one post-body source functional. The construction is unique at the level of physical roles, modulo relocation of a finite local counterterm between the geometric and quantum bookkeeping columns while holding their sum fixed.*

Proof. Functional differentiation of one W_O^{ren} gives the four displayed terminal owners. Diffeomorphism and gauge invariance give their common Ward covector. The fluctuation Green operator is the inverse of the corresponding gauge-fixed Hessian and therefore computes response; inserting it again as an action summand would duplicate the fluctuation contribution. A finite local counterterm can be written in either the geometric or renormalized quantum column, but linearity of variation shows that only the sum affects the field equations. Including the same term in both columns doubles its variation. Thus the role ledger is fixed, while its harmless counterterm placement convention is not. $\square$

Scope. Theorem 1 is a construction theorem inside the selected ordered completion. It does not show that the physical base-seed law selects Γ_{PC} , $\mathcal{C}_O$, its field spectrum or its renormalized continuum measure.

The ordering is structural. If $H_B, H_P > 0$ and the post-body equations have body derivative C , the linear response is lower triangular:

$$\begin{pmatrix} \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} H_B^{-1} & 0 \\ -H_P^{-1}CH_B^{-1} & H_P^{-1} \end{pmatrix} \begin{pmatrix} \delta j_B \\ \delta j_P \end{pmatrix}. \quad (8)$$

Thus morphology conditions the physical terminals, while a later terminal load does not rebuild the body.

Proposition 1 (Simultaneous-action no-go). *For an invertible positive simultaneous Hessian*

$$\mathbb{H} = \begin{pmatrix} H_B & C^\dagger \\ C & H_P \end{pmatrix}, \quad (9)$$

exact vanishing of body response to every pure post-body load implies $C = 0$. Hence nonzero body conditioning and exact body freeze cannot both be encoded by one simultaneous unconstrained Hessian solve.

Proof. Schur inversion gives $\delta y = -H_B^{-1}C^\dagger(H_P - CH_B^{-1}C^\dagger)^{-1}\delta j_P$. If this vanishes for every δj_P , invertibility forces $C = 0$. $\square$

5 Disjoint occupation and stress ownership

Let the occupied mode labels form a disjoint union

$$\mathcal{I}_{\text{occ}} = \mathcal{I}_{\text{cl}} \sqcup \mathcal{I}_q. \quad (10)$$

Classical coherent modes represented in Γ_{occ} are excluded from the state-dependent part of Γ_q ; vacuum polarization and local renormalization terms occur only in $\Gamma_q + \Gamma_{\text{ct}}$. This disjoint partition is a sufficient transparent ledger, not the only possible field-theoretic organization. A background-field, one-particle-irreducible or two-particle-irreducible construction is also admissible when its mean field, fluctuation determinant and counterterms are derived from one effective action and no full contribution is inserted twice.

Proposition 2 (Duplicate-stress no-go). *If one nonzero occupied mode i is included in both $\mathcal{I}_{\text{cl}}$ and $\mathcal{I}_q$ with the same metric variation, its stress contribution is counted twice. No terminal redefinition removes this without changing the metric equation or adding a compensating subtraction.*

Proof. Metric variation is linear. The duplicated summands give $-2(\sqrt{-g})^{-1}\delta(\Gamma_i + \Gamma_i)/\delta g^{\mu\nu} = 2T_{\mu\nu}^{(i)}$. A subtraction merely moves the ownership entry and must therefore be recorded explicitly in Γ_{ct} . $\square$

6 Joint terminal theorem

Assume on the selected Lorentzian leaf

$$\Gamma_{\text{grav}} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} (R - 2\Lambda), \quad (11)$$

$$\Gamma_{\text{EM}} = -\frac{Z_F}{4} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}, \quad Z_F = p_* K_Y, \quad (12)$$

and let the matter and renormalized quantum functionals be diffeomorphism and $U(1)$ gauge invariant. These are standard-limit forms inherited conditionally; Paper VII does not derive their Nature-level selection.

Theorem 2 (Common-descent Einstein–Maxwell–matter assembly). *Under the ordered ledger, disjoint occupation rule, common unit packet and the stated invariances, stationarity of Eq. (4) gives*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(T_{\mu\nu}^{\text{EM}} + T_{\mu\nu}^{\text{cl}} + \langle T_{\mu\nu}^q \rangle_{\text{ren}} \right), \quad (13)$$

$$\nabla_\mu (Z_F F^{\mu\nu}) = J_{\text{cl}}^\nu + \langle J_q^\nu \rangle_{\text{ren}}. \quad (14)$$

On the matter and gauge equations, the Ward identities and Bianchi identity imply

$$\nabla^\mu T_{\mu\nu}^{\text{tot}} = 0, \quad \nabla_\mu J_{\text{tot}}^\mu = 0. \quad (15)$$

No additional gravity, gauge or quantum-backreaction gain is introduced at the terminal stage.

Proof. Equations (13) and (14) are the metric and gauge Euler equations of the single post-body functional. Gauge invariance gives current conservation. Diffeomorphism invariance gives the combined stress Ward identity, including the Lorentz-force exchange between matter and gauge sectors. The divergence of the left side of Eq. (13) vanishes identically, so the equations are mutually consistent. Disjoint occupation and the unique counterterm ledger ensure that each contribution is varied once. $\square$

Corollary 1 (Standard weak-field leaf). *For $g_{00} = -(1 + 2\Phi_N/c^2)$, stationary weak fields and nonrelativistic sources, Eq. (13) reduces to*

$$\nabla^2 \Phi_N = 4\pi G \rho_{\text{tot}}, \quad (16)$$

where ρ_{tot} is the same once-counted density appearing in the ledger. Equation (14) reduces to the ordinary Maxwell equations after the inherited field normalization. This is standard recovery, not a Tau-specific deviation.

7 Local constraint structure

The covariant equations must reproduce constraints, not only evolution equations. Let n^μ be the unit normal to an observer-compatible spacelike leaf, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ its induced metric, and K_{ij} its extrinsic curvature. With $\rho = n^\mu n^\nu T_{\mu\nu}^{\text{tot}}$ and $j_i = -h_i^\mu n^\nu T_{\mu\nu}^{\text{tot}}$, the normal projections of Eq. (13) give, in the stated sign convention,

$${}^{(3)}R + K^2 - K_{ij}K^{ij} = 2\kappa\rho + 2\Lambda, \quad (17)$$

$$D_j(K^{ij} - h^{ij}K) = \kappa j^i. \quad (18)$$

The normal projection of Eq. (14) gives the Abelian Gauss constraint

$$D_i(Z_F E^i) = \rho_J. \quad (19)$$

The Ward identities propagate these constraints when the classical and quantum matter equations hold. Paper VII therefore imports the standard local constraint architecture from the one joint functional; it does not derive a new constraint algebra or quantize it.

8 Primitive terminal inventory

The assembled packet contains several typed outputs, not several names for the same information:

Family	Primitive terminal content	Derived or composite descendants
Geometry/gravity	g , connection, curvature and constraints	proper time, tidal response, lensing after a null source
Gauge/radiation	A , F , conserved current, two screen polarizations	photon propagation, redshift and optical transport
Matter occupation	occupied localized or propagating solutions Φ	mass, energy, stress and charge records
Quantum operation	ρ , effects, instruments and correctable composition	Born statistics and renormalized quantum stress/current
Observer access	causal orientation, clock and stable restriction	observer-relative terminal classes

Stress-energy, current, redshift and lensing are variational or composite outputs of these families; they are not silently counted as new primitive source directions. Conversely, the symmetric local gravity descriptor cannot stand in for internal gauge charge, quantum phase or occupied matter. A global information-completeness claim must wait for the non-Abelian and fermionic known-terminal completions.

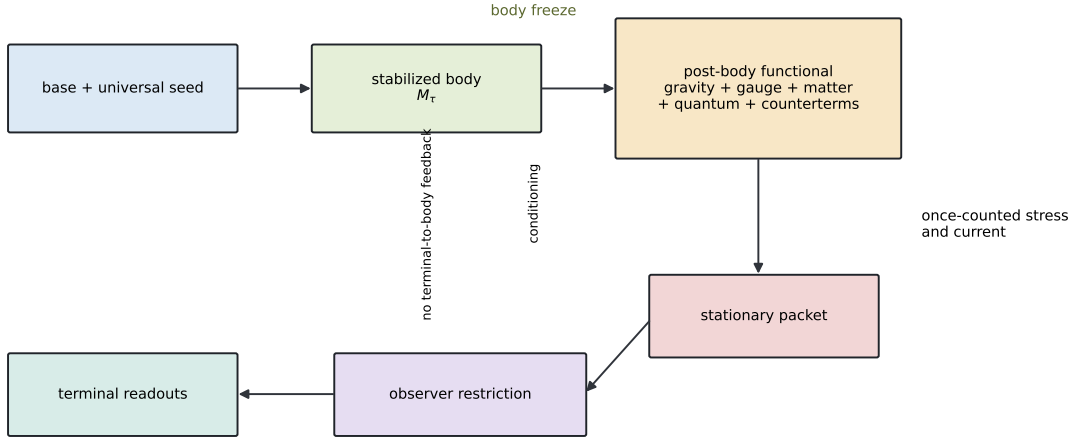


Figure 1: Ordered common-source construction. The body is stabilized before the post-body functional is solved; observer restriction is terminal.

9 Why the standard limit does not prove common provenance

Proposition 3 (Equal-terminal provenance no-go). *The equations (13)–(15) do not imply that their coefficients or sectors arise from one Tau source.*

Proof. For any nonzero constants a_G, a_F, a_M, a_q , independently rescale the four sector functionals and compensate by redefining their terminal couplings so that the displayed equations are unchanged. The terminal solution set is the same while the source ownership graph and common-unit identification differ. Therefore standard-limit agreement cannot be inverted into common provenance. $\square$

This no-go fixes the empirical interpretation: terminal data may test a source-frozen forward ledger, but may not choose its source partition, susceptibilities or unit map.

Proposition 4 (Same-reduct selection no-go). *The currently frozen stationary body, common action line, observer quotient and terminal type signatures do not entail physical occupation of the full effective-source packet of Theorem 1.*

Proof. Hold fixed the body y_* , its Hessian, common units, causal orientation and observer quotient. Completion C_0 assigns zero occupation to one admissible post-body record/field edge. Completion C_1 assigns a nonzero positive occupation to that same typed edge and includes its contribution in Γ_{PC} . Both have the same frozen upstream reduct and satisfy its positivity and typing conditions, but only C_1 realizes the corresponding probability, stress and current derivatives in Eqs. (6)–(7). Hence the upstream reduct does not select the full packet. Additional source-owned occupation or an empirically discriminating source-frozen prediction is required. $\square$

This finite counterpair is the stopping result for the origin question at the present source level. It forbids both a Nature-level promotion of Theorem 1 and another same-level search for a hidden normalization. It does not weaken the conditional forward construction.

10 Quantum backreaction and the quantum-gravity boundary

The quantum term in Eq. (13) is a renormalized expectation value on a classical body-derived metric, in the standard semiclassical sense [Birrell and Davies, 1982, Wald, 1977]. Local covariance and admissible renormalization follow the usual curved-spacetime QFT framework [Hollands and Wald, 2001, Brunetti et al., 2003]. The metric can also be driven by a stress-noise second cumulant in a stochastic extension, but this does not quantize geometry.

Paper VII therefore establishes a conditional quantum-gravity *interface*, not quantum gravity. A path integral or operator algebra for the metric, gravitational gauge fixing, a physical inner product and a nonperturbative ultraviolet completion remain absent.

11 Finite decision ledger

Question	Verdict	Status
One simultaneous body/post-body Hessian supplies conditioning and freeze	Impossible unless the cross derivative vanishes	Exact no-go
One ordered ledger supplies both	Yes, Eq. (8)	Conditional theorem
One post-body effective functional generates probability, stress, current and clock response	Yes, Eqs. (5)–(7)	Conditional theorem
Green response is an additional action contribution	No; it is an inverse Hessian	Exact bookkeeping result
Finite local counterterms may be included in two columns	No; relocation is allowed, duplication is not	Exact bookkeeping result
Classical and quantum occupation may overlap silently	No; stress doubles	Exact no-go
Einstein, Maxwell and Ward equations share one post-body variation	Yes under the declared packet	Conditional theorem
Newton/Maxwell limits follow	Yes on the standard weak leaf	Conditional recovery
Standard-limit agreement proves Tau provenance	No	Exact no-go
Geometry is quantized	Not derived	Open
Hamilton, momentum and Gauss constraints follow	Yes, on the selected standard leaf	Conditional recovery
Non-Abelian gauge and Standard Model spectrum follow	Not derived	Open
Physical base-seed source selects $\mathfrak{L}_7$	Not entailed by the current reduct, Proposition 4	Exact current-reduct no-go

12 Conclusion

Paper VII closes a specific assembly problem. The inherited primitive terminals can be placed in one ordered, once-counted effective functional without rebuilding the morphology from down-

stream loads. Diffeomorphism and Abelian gauge invariance then make gravity, current and stress conservation one consistent variational packet, while the disjoint occupation rule prevents a classical/quantum duplicate source. The Newtonian and Maxwell limits are recovered from that same packet.

The construction remains conditional, but its remaining source status is no longer an unspecified proof gap. An explicit same-reduct counterpair proves that present Tau source laws do not entail occupation of the complete ledger. Progress therefore requires a richer independently motivated source law or a source-frozen cross-terminal prediction. Terminal fitting cannot supply the missing selection. Absolute constants, quantized geometry, non-Abelian structure and the particle spectrum remain outside the present result.

References

- N. D. Birrell and P. C. W. Davies. *Quantum Fields in Curved Space*. Cambridge University Press, 1982. doi: 10.1017/CBO9780511622632.
- Robert M. Wald. The back reaction effect in particle creation in curved spacetime. *Communications in Mathematical Physics*, 54:1–19, 1977. doi: 10.1007/BF01609833.
- Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001. doi: 10.1007/s002200100540.
- Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch. The generally covariant locality principle: A new paradigm for local quantum physics. *Communications in Mathematical Physics*, 237: 31–68, 2003. doi: 10.1007/s00220-002-0722-0.